

FIFA TRANSFERCHAIN

Settlement Engine for Football Player Transfers

An on-chain protocol for transfer fees, solidarity contributions, sell-on cascades, and agent commissions

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Live walkthrough on Sepolia and team contributions

SECTION 01

Introduction & Business

Problem framing and the case for blockchain settlement

What is TransferChain?

A blockchain-based settlement protocol that automates every financial obligation embedded in a professional football transfer — released on a pre-encoded installment schedule.

Transfer Fee

The principal payment from buyer to seller, released in scheduled installments — typically 2–5 tranches over multiple seasons.

Solidarity

5% of every installment, distributed to every training club that developed the player (FIFA RSTP Art. 21).

Sell-on cascade

Pre-encoded percentage payments to prior owners, automatically split at every release.

Agent commission

Up to 6% per agent, hard-capped on-chain to enforce FIFA's commission ceiling.

All four streams settled in a single self-executing smart contract, on the schedule clubs actually negotiate.

Why Blockchain?

The problem isn't the rules. It's the enforcement.

FIFA already mandates solidarity contributions and commission caps. Compliance fails because no party has the standing or the visibility to make the rules stick.

01 Enforced by code

Once entitlements are written on-chain, no party can override or selectively exclude them.

02 Multi-party attestation

FIFA, the national association, and the buyer club must all sign before activation.

03 Time-locked release

Payments trigger automatically once `block.timestamp ≥ dueDate`. No follow-up required.

Goal: meaningful trust minimization, not full trustlessness.

“

The system has encouraged exorbitant fees, secretive financial transactions and repeatedly fails players.

FIFPro — World Players' Union in Football

#GameChangers campaign · gamechangers.football

Present Problems

Two failure modes drive the dispute volume.

DELAYED & UNPAID

19.3%

of solidarity contributions

actually paid in 2018, before the FIFA Clearing House launched

African and South American academies that develop top-tier players regularly receive nothing.

DISPUTED & LITIGATED

21,633

cases at the FIFA Football Tribunal

in 2024/25 — a record

Brighton spent two years at FIFA and CAS recovering €187,500 after Málaga structured a player's exit to dodge a sell-on clause.

The Solution

A single self-executing contract replaces a fragmented, trust-dependent settlement process.

01

Automated Distribution

Every recipient — academy, agent, sell-on club — is registered on-chain at contract creation. Payments push automatically when $\text{block.timestamp} \geq \text{dueDate}$.

No one needs to remember. No one can refuse.

02

No Room for Dispute

All distribution shares are locked in basis points at deployment. The math is transparent and immutable.

There is no ambiguity to litigate — only code to execute.

How TransferChain Compares

STATUS QUO



Solidarity payments only

Sell-on clauses and agent commissions out of scope of any centralized clearing.



Centralized & opaque

Academies have no visibility into the calculation or status of their payments.



Trust-dependent

Clubs must rely on FIFA to identify, calculate, and process payments correctly.

TRANSFERCHAIN



All four streams unified

Transfer fee, solidarity, sell-on, and agent commissions settled in one contract.



On-chain & transparent

Any party can verify their entitlement and watch funds settle in real time.



Permissionless release

Anyone can trigger the release once due date passes — no intermediary required.

Competitive Landscape

No existing solution unifies all four payment streams in self-executing infrastructure.

	Scope	Architecture	Status
FIFA Clearing House	Solidarity & training rewards only	Centralized, regulated payment institution	Live since 2022; >\$350M distributed
WIN × MANTRA	Investment product on top of solidarity	Tokenized revenue streams	~\$7M AUM; not settlement
Academic proposals	Theoretical blockchain integration	Whitepapers in IEEE, Nature	No live implementation
TransferChain	All four payment streams	Self-executing smart contract	Sepolia-ready prototype

TransferChain is the only system that automates settlement of every transfer obligation in a single contract.

SECTION 02

Smart Contract

On-chain enforcement of FIFA payment rules

Smart Contract

01. Contract Lifecycle

The contract moves through four phases — from deployment to cascading payout — with every step enforced on-chain.

01 Deploy

Constructor seeds roles, fee, and deal name.

02 Activate

FIFA, association, and buyer must all sign (3-of-3).

03 Register

Recipients and installments encoded on-chain.

04 Release

Time-locked payouts cascade to all parties.

Next: zooming into four implementation details that make this work.



Smart Contract

02. FIFA Rules as Constants

Compliance rules aren't checked manually — they're immutable constants enforced at every transaction.

Solidarity rate (5%) and agent commission cap (6%) are hard-coded; no party can re-negotiate them at runtime.

```
// Solidarity contribution fixed at 5% – FIFA RSTP Art. 21
uint16 public constant SOLIDARITY_RATE_BPS = 500;

// Agent commission hard cap at 6% – FIFA regulations
uint16 public constant AGENT_CAP_BPS = 600;

// All recipient types locked at deployment
enum RecipientType {
    SELLING_CLUB,      // counterparty selling the player
    SOLIDARITY_CLUB,   // training academy (FIFA RSTP Art. 21)
    SELLON_CLUB,       // club holding a sell-on clause
    AGENT,             // licensed intermediary
    PROTOCOL_FEE       // TransferChain protocol wallet
}
```

Smart Contract

03. 3-of-3 Consensus Gate

No single party can activate the contract alone. FIFA, the national association, and the buyer club must all sign before any funds can be deposited.

Three independent approvals are required to flip the contract from AWAITING_CONSENSUS to ACTIVE — eliminating any unilateral attack vector.

```
function approveActivation() external onlyAwaitingConsensus {
    bool isFifa = hasRole(FIFA_ADMIN_ROLE, msg.sender);
    bool isAssoc = hasRole(NAT_ASSOC_ROLE, msg.sender);
    bool isBuyer = hasRole(BUYER_CLUB_ROLE, msg.sender);

    require(isFifa || isAssoc || isBuyer, "Not an authorized approver");
    if (hasApproved[msg.sender]) revert AlreadyApproved();

    hasApproved[msg.sender] = true;
    approvalCount++;

    // Activates only when all 3 approvals are received
    if (approvalCount >= REQUIRED_APPROVALS) {
        status = ContractStatus.ACTIVE;
    }
}
```

Smart Contract

04. Time-Lock & Auto Distribution

Anyone can call once the due date is reached — no intermediary required. A failed push doesn't block the rest; it's logged for admin recovery.

Each due installment is released to all recipients in one transaction, with payout failures isolated and recoverable rather than blocking subsequent releases.

```
function releaseNextInstallment() external onlyActive nonReentrant {

    // Time-lock: reverts if installment not yet due
    if (block.timestamp < inst.dueDate)
        revert InstallmentNotDue();

    // CEI pattern: state updated before transfers (re-entrancy-safe)
    inst.status = InstallmentStatus.RELEASED;
    nextInstallment++;

    // Dust-safe: integer-division remainder added to first entry
    payments[0] += inst.amount - totalCalc;

    // Single transaction pushes funds to all recipients
    for (uint8 i = 0; i < len; i++) {
        (bool sent, ) = rec.wallet.call{value: payments[i]}("");
        if (!sent) failedPayments[idx][rec.wallet] += payments[i];
    }
}
```

Smart Contract

05. On-Chain Rule Validation

Invalid distributions are rejected at registration, not after the fact. Overpaid agent commissions and incomplete splits cannot enter the contract in the first place.

Distribution shares are validated against FIFA limits and must sum to exactly 100% — the contract refuses to deploy with invalid splits.

```
function _validateDistribution(
    uint8[] calldata idxs,
    uint16[] calldata bps
) internal view {
    uint256 total = 0;
    for (uint8 i = 0; i < idxs.length; i++) {

        // Revert if any agent entry exceeds the 6% cap
        if (recipients[idxs[i]].recipientType == RecipientType.AGENT)
            if (bps[i] > AGENT_CAP_BPS)
                revert AgentCommissionExceedsCap();

        total += bps[i];
    }

    // Revert if shares do not sum to exactly 100%
    if (total != BASIS_POINTS) revert InvalidBasisPoints();
}
```


SECTION 03

Business & Market

Target customers, market size, and go-to-market

Target Market

Four customer segments capture every party in a typical transfer settlement.

PROFESSIONAL FOOTBALL CLUBS

Selling Clubs

Pay the protocol fee (10–20 bps) to guarantee the security and punctuality of incoming receivables.

PROFESSIONAL FOOTBALL CLUBS

Buying Clubs

Use the platform to automate outward payments across solidarity, sell-on, and agent obligations.

STAKEHOLDERS

Training Clubs

Small-to-medium academies entitled to FIFA Solidarity Contributions — receive funds automatically.

STAKEHOLDERS

Football Agents

Licensed intermediaries paid commissions automatically, capped at the 6% FIFA ceiling.

Market Size

Modest adoption of a 10–20 bp protocol fee on a \$13B annual market is meaningful revenue.

TAM

\$13.1B

Total Addressable Market

2025 spending on international football transfer fees
(FIFA Global Transfer Report 2025).

SAM

~\$10.5B

Serviceable Addressable Market

~80% of professional transfers involve installments,
sell-on clauses, or solidarity payments.

SOM

~\$131M

Serviceable Obtainable Market

Year 1 target — capturing 1% of global volume settled
through the protocol.

Go-to-Market Strategy

Domestic-first, then cross-border — solving the network-effect problem one league at a time.

2026–2027

SINGLE-LEAGUE PILOT

- ▶ Launch in one domestic league (Eredivisie or Liga Portugal — strong export markets, single national association as oracle).
- ▶ Scope: domestic transfers only — single-jurisdiction settlement avoids cross-border verification gaps.
- ▶ Sepolia testnet, then mainnet with first cohort of clubs.

2027–2028

MULTI-LEAGUE EXPANSION

- ▶ Onboard additional domestic leagues — still domestic-only — to build network breadth.
- ▶ Integrate with FIFA Clearing House data standards as a complementary settlement rail.
- ▶ Onboard top-tier agencies to automate commission payouts.

2028+

CROSS-BORDER ACTIVATION

- ▶ Once enough national associations are onboarded, enable cross-border transfers between participating leagues.
- ▶ Introduce receivable financing — clubs discount future installments via the protocol.
- ▶ Layer-2 deployment for lower transaction costs at scale.

SECTION 04

Demo & Team

Live walkthrough on Sepolia and team contributions

Product Demo

End-to-end walkthrough on the Sepolia testnet — five steps, one transfer.

STEP 01

Deploy

Constructor seeds roles, protocol fee, and deal name.

STEP 02

Activate

FIFA admin, association, and buyer club each call `approveActivation()`.

STEP 03

Register

FIFA admin registers recipients and the installment schedule.

STEP 04

Deposit

Buyer club deposits the full transfer fee into escrow.

STEP 05

Release

Time-locked release pushes funds to all recipients in one transaction.

Cascading payouts to all five recipients (seller, solidarity, sell-on, agent, protocol) settle in a single transaction.

Watch the Demo

Recorded walkthrough of the full transaction flow on Sepolia.



TransferChain — End-to-End Demo

Sepolia testnet · ~5 minutes · five steps

https://drive.google.com/file/d/1NIuh0iFylT_p6gzNRhgAgpDHQNCp-zhd/view

Team & Contributions

MK

Maximilian Kunz

Business Strategy & Lifecycle Design

Led business plan, contract architecture, lifecycle diagram, and end-to-end deck design.

YL

Yan Lu

Smart Contract Engineering

Solidity implementation, OpenZeppelin integration, Sepolia deployment, and unit testing.

YJ

Yiheng Jiang

Product & Demo Production

Frontend integration with MetaMask via ethers.js; recorded and edited the demo walkthrough.

YH

Yun-Chien Huang

Market Analysis & GTM

Competitor research, market sizing, and the three-phase go-to-market strategy.

Questions?

Thank you for your attention.

FIFA TRANSFERCHAIN · GROUP 14

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